

A Gaussian subcase of the Veraar–Weis Theorem 4.2 tail question

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Source and target

The source paper is Veraar and Weis, *A note on maximal estimates for stochastic convolutions*, arXiv:1004.5061 [1]. The paper proves maximal estimates for stochastic convolutions

$$S \diamond G(t) = \int_0^t S(t-s)G(s) dW_H(s)$$

under an H^∞ -calculus assumption on the generator. The first exponential-tail question in the paper is answered immediately by Theorem 2.4 for UMD spaces of type 2. The surviving open target is the paragraph after Theorem 4.2:

At this moment we do not know if there are exponential tail estimates in the general setting of Theorem 4.2. However, also in this setting there is some hope that $C_{p,Y}^\gamma \leq C_X \sqrt{p}$ for p large, and by the argument in Theorem 2.4 this would yield exponential tail estimates again.

The same paragraph records that Cox–Veraar obtained only $C_{p,Y}^\gamma \lesssim_X p$, which gives exponential estimates but not the expected quadratic exponential estimates.

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Result

The full adapted problem remains open, but the quadratic tail holds in the deterministic and initially independent subcase of Theorem 4.2.

Theorem 1. *Let X be a UMD Banach space with property (α) , and let A satisfy hypothesis (H) of [1]. Let $S(t) = e^{tA}$. Let $(W_H(t))_{t \geq 0}$ be an H -cylindrical Brownian motion, and suppose that the filtration has the form*

$$\mathcal{F}_t = \mathcal{G} \vee \sigma(W_H(s) : 0 \leq s \leq t),$$

where $\mathcal{G}$ is independent of W_H . Let

$$G : \Omega \rightarrow \gamma(\mathbb{R}_+; H, X)$$

be $\mathcal{G}$ -measurable and suppose that for some $M > 0$,

$$\|G(\omega)\|_{\gamma(\mathbb{R}_+; H, X)} \leq \sqrt{M} \quad \text{for a.e. } \omega.$$

Then $S \diamond G$ has the continuous modification supplied by Theorem 4.2, and if $K_2 = K_2(A, X)$ denotes the $p = 2$ maximal-estimate constant in Theorem 4.2, then for every $u \geq 0$,

$$\mathbb{P}\left(\sup_{t \geq 0} \|S \diamond G(t)\| \geq K_2 \sqrt{M} + u\right) \leq \exp\left(-\frac{u^2}{2K_2^2 M}\right).$$

Consequently, for every $\lambda \geq 2K_2 \sqrt{M}$,

$$\mathbb{P}\left(\sup_{t \geq 0} \|S \diamond G(t)\| \geq \lambda\right) \leq \exp\left(-\frac{\lambda^2}{8K_2^2 M}\right).$$

Proof Intuition

The obstruction in the adapted case is that the integrand can react to the Brownian path; moment estimates then pass through Banach-space stochastic integral constants. If the integrand is known at time zero and independent of the Brownian motion, the obstruction disappears. Conditioning on $\mathcal{G}$ freezes G , and the stochastic convolution becomes a centered Gaussian process indexed by $t \geq 0$.

Theorem 4.2 at $p = 2$ gives an L^2 bound for the supremum of that process. For a centered Gaussian process, this same bound controls both the mean scale and the weak variance parameter. Borell's inequality then upgrades the L^2 maximal estimate to a quadratic tail.

Proof

Fix a deterministic $g \in \gamma(\mathbb{R}_+; H, X)$. Theorem 4.2 of [1], with $p = 2$, gives the continuous modification and the estimate

$$\left(\mathbb{E}_W \sup_{t \geq 0} \|S \diamond g(t)\|^2\right)^{1/2} \leq K_2 \|g\|_{\gamma(\mathbb{R}_+; H, X)}, \quad (1)$$

where $\mathbb{E}_W$ denotes expectation over the Brownian motion and K_2 depends only on the constants in Theorem 4.2 for A and X .

For deterministic g , the process $Z_t = S \diamond g(t)$ is centered Gaussian. Since the modification is continuous, the random variable

$$Z^* = \sup_{t \geq 0} \|Z_t\|$$

is the supremum of a separable centered Gaussian process after restricting time to the rational points and using a countable norming set for the separable closed span of the path. Its weak variance parameter satisfies

$$\sigma_g^2 = \sup_{t \geq 0, \|x^*\| \leq 1} \mathbb{E}_W |x^* Z_t|^2 \leq \mathbb{E}_W (Z^*)^2 \leq K_2^2 \|g\|_{\gamma(\mathbb{R}_+; H, X)}^2$$

by (1). Also

$$\mathbb{E}_W Z^* \leq (\mathbb{E}_W (Z^*)^2)^{1/2} \leq K_2 \|g\|_{\gamma(\mathbb{R}_+; H, X)}.$$

Borell's inequality for separable centered Gaussian processes therefore gives

$$\mathbb{P}_W(Z^* \geq K_2 \|g\|_{\gamma(\mathbb{R}_+; H, X)} + u) \leq \exp\left(-\frac{u^2}{2K_2^2 \|g\|_{\gamma(\mathbb{R}_+; H, X)}^2}\right)$$

for $u \geq 0$, with the evident trivial interpretation when $g = 0$.

Now return to the random initially independent integrand G . Conditional on $\mathcal{G}$, the preceding deterministic argument applies to $g = G(\omega)$. Using $\|G(\omega)\|_{\gamma(\mathbb{R}_+; H, X)} \leq \sqrt{M}$ gives, a.s.,

$$\mathbb{P}\left(\sup_{t \geq 0} \|S \diamond G(t)\| \geq K_2 \sqrt{M} + u \mid \mathcal{G}\right) \leq \exp\left(-\frac{u^2}{2K_2^2 M}\right).$$

Taking expectation in $\mathcal{G}$ proves the first displayed estimate. If $\lambda \geq 2K_2 \sqrt{M}$, put $u = \lambda - K_2 \sqrt{M}$; then $u \geq \lambda/2$, which gives the second estimate.

Scope and verification notes

This packet gives a solved subcase, not a full answer to the source paper's general adapted question. The proof uses Gaussianity after conditioning on the initial sigma-field. For a genuinely adapted integrand there is no such conditional Gaussian process, and the source paper's proposed moment route still requires the open $\sqrt{p}$ -growth of the relevant UMD stochastic-integral constants.

The boundary is consistent with later literature. Cox and Veraar [2, p. 28] prove p -independence phenomena for decoupling, but explicitly state that the optimal $\kappa_{p,X}/\sqrt{p}$ boundedness is open already for $X = L^q$, $q \neq 2$. Later general UMD Burkholder–Davis–Gundy theorems, such as Yaroslavl'tsev [3], provide qualitative Banach-valued BDG estimates but do not remove this $\sqrt{p}$ constant obstruction for the adapted Theorem 4.2 tail problem.

Local cheap-index search found no existing result packet for arXiv:1004.5061. The exact source paragraph, $C_{p,Y}^\gamma$, Cox–Veraar arXiv:1107.2218, and Yaroslavl'tsev arXiv:1807.05573 were checked as bounded literature/status search terms. No full adapted quadratic-tail answer was found in that check.

Human-review recommendation

Review the separability step in the application of Borell's inequality and the conditional-use formulation for initially independent integrands. The mathematical dependence on the source theorem is only the $p = 2$ maximal estimate from Theorem 4.2, so the result should be robust once those standard Gaussian-process details are accepted.

References

- [1] M. Veraar and L. Weis. *A note on maximal estimates for stochastic convolutions*. arXiv:1004.5061, 2010.
- [2] S. Cox and M. Veraar. *Vector-valued decoupling and the Burkholder–Davis–Gundy inequality*. arXiv:1107.2218, 2011.
- [3] I. Yaroslavl'tsev. *Burkholder–Davis–Gundy inequalities in UMD Banach spaces*. arXiv:1807.05573, 2018.
- [4] M. Ledoux. *The concentration of measure phenomenon*. American Mathematical Society, 2001.